NUIMPACT

Sector Spotlight Digest

Fall 2024 | Volume I

Northeastern University

NUIMPACT
IMPACT INVESTING INITIATIVE

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NUImpact Overview

About NUImpact

- NUImpact is Northeastern University's student-led impact investing initiative and investment fund.
- Since our inception in Spring 2016, the organization has served as a unique resource and thought exchange for the Northeastern community to understand purposeful capital, develop technical skills, and make targeted investments in companies that promote positive social and environmental impact
- Our three-pronged investment thesis seeks investment in companies that are:

Northeast-Focused | **Aid Underserved Communities** | **Financially Sustainable**

Our Research Arm Yields Significant Benefits

- The research analyst program places a strong emphasis on industry research to enhance our fund's impact diligence process.
- A key responsibility for research analysts is crafting concise one-page reports spotlighting a chosen sector within their verticals.
- Some benefits of this new program include:

Informed Decision Making

Offers the fund with valuable insights into specific industries, enabling risk mitigation and better-informed investment decisions

Refined Impact Assessment

Enhanced sector understanding equips analysts with knowledge needed to conduct robust impact diligence and measurement

Industry Focus

- **Education & Financial Services**
- **Energy & Environment**
- **Food & Agriculture**
- **Healthcare**
- **Technology, Media, & Telecommunication**

Meet the Research Team

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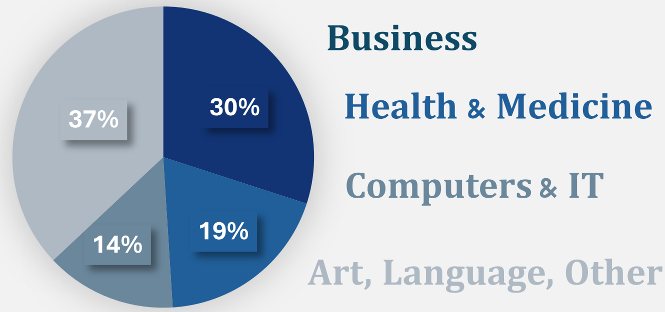
Education & Finance: Online Life-Long Learning

Industry Overview

- Online courses give professional and developing workers an easy resource to grow and build skills.
- Platforms, such as Freemium models and massive open online courses(MOOCs), offer partially or entirely free courses, which allow for ease of use and low initial investment by the user
- Web-based trainings offer a solution to many topics missed within primary, secondary, and university-level schooling.

Course Subject Breakdown



Future Outlook

Tailwinds

Increased demand for **technical skills** accessible through online courses

Lowering prices and free MOOCs make courses **highly accessible**

21st Century Distance Education Guidelines promoting higher quality online education courses

Headwinds

Over-competition and **market competitiveness** lowered the **validity** of each individual course

Low completion rate among certain courses, exponentially more common among longer courses

Changing professional landscape quickly **outdating** and **outclassing** certain courses

Policy Spotlight Overview

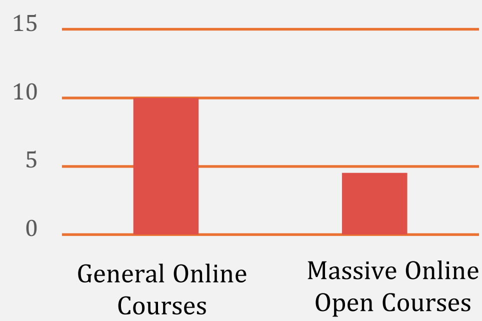
New England Adult Education Virtual School

- Pre-Secondary education training to New England adults without a high school diploma

Assisting young workers gain advanced academic skills necessary for employment

66% of institutions offer fully online programs and 46% offer hybrid options

Completion Rate Among Course Types



Recent Deals Highlights

CORP/U

Acquired by Udemy

Augmenting online offerings to become cohort-based through Crop U's highly sociable platform

POINTFUL EDUCATION

Acquired by Savvas Learning Co.

Enhancing professional online development offerings for primary and secondary students

NEW ENGLAND COLLEGE of BUSINESS

Acquired Cambridge College

Expansion of Cambridge Colleges online offerings for both students and non-student users

Energy & Environment: Energy Storage

Industry Overview

Market Description

- Energy storage systems collect energy when it's produced, enabling it to be used later to balance potential gaps between supply and demand.
- Scaling is crucial for net-zero goals and enabling widespread renewable energy deployment.

21% CAGR → 137GW/442GWh by 2030

Current Industry Trends

- **Advanced Thermal Energy Storage** uses mediums such as molten salts to store renewable heat, balancing supply and demand within solar and heating systems.
- **Distributed Storage Systems** enable on-site energy production and storage to reduce grid failure risks and address supply chain disruptions during high demand.

Industry Tailwinds

2022 Inflation Reduction Act

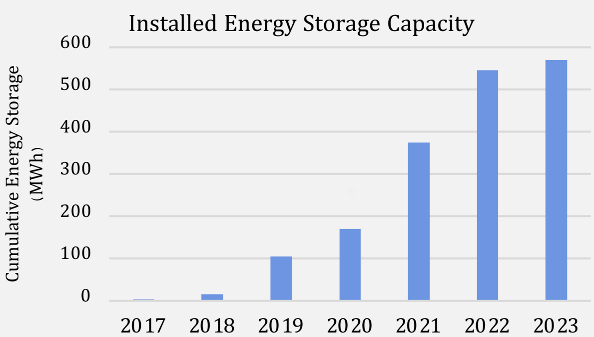
- Allocated $369 billion for climate initiatives, including expanding the 30% Investment Tax Credit (ITC) to standalone energy storage.

Incentives energy storage development

Stimulate research and development

Optimizes Economic Returns

Massachusetts Energy Storage Initiative (ESI)



- 569 MWh of installed energy storage was reported at end of 2023, 57% of the way to ESI's 1000 MWh goal by end of 2025.

Power Up New England Proposal

$389M Of federal funding to New England States for **Transmission** and **Energy Storage** Infrastructure

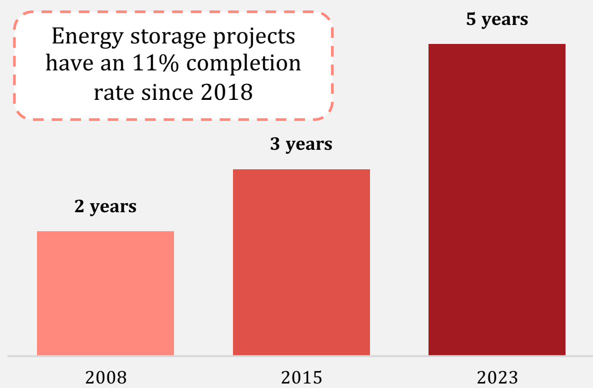
- Centers around investments in development of offshore wind and battery energy storage opportunities.

Industry Headwinds

Backlogged Interconnection Queues

- ~1,030 Gigawatts of projects waiting for grid connection **hinder new storage deployments** and limit meeting stark, rising demand

Average Time Spent in Interconnection Queues



Lack of Standardization

- The lack of standardized specifications for energy storage systems impedes innovation, raises costs, and slows adoption of technologies in the sector.

Company Spotlight

Founded: 2020

florrent

Round:

August 12th, 2024, Florrent successfully closed $3.6 million in Seed funding led by **MassVentures**.

Company Description:

Florrent Inc. develops **bio-derived carbon materials** for **backup power systems**, with applications in **renewable energy** and **grid stability**, offering efficient and sustainable energy storage.

Food & Agriculture: Green Ammonia Production & Fertilizer

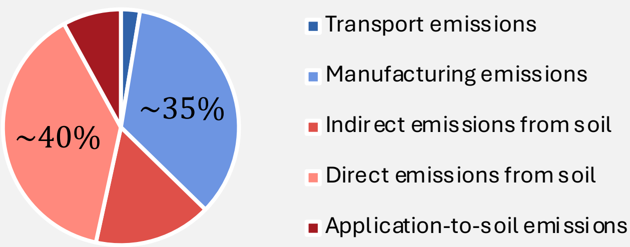
Industry Overview

- The industry has a current valuation of $0.3B

66% CAGR → **$6.2** Billion Market Size by 2030

- Over **400 large-scale green Hydrogen and Ammonia projects** have recently begun, likely to be **followed by smaller regional-scale projects**

Estimated Global Greenhouse Gas Emissions from Nitrogen Fertilizer



Traditional vs. Green Fertilizer

- **Production of traditional fertilizer** accounts for **70%** of ammonia production, where **2 tonnes of emissions are released** per **1 ton output** of ammonia
- The **green production of ammonia and fertilizer releases less emissions,** as it can be **produced on farms** or **locally sourced**

Traditional Fertilizer		Green Fertilizer	
Produced in **500 large-scale facilities** worldwide		Can be **produced on-farm or** at a **larger scale,** with locally sourced **solar panels, green hydrogen storage**	
Outdated& **energy intensive**	The process is **responsible** for **1.3% of global greenhouse gas emissions,** per year (Aviation is 2%....)	Can **reduce** Agricultural Products' **greenhouse gas emissions by 5%**	
It feeds **half** of the **global population,** alone		By 2030, **96% of the current industry** can be **at scale** this way	
Caveats			
Severe **Supply Chain risks**	Severe **price increase vulnerabilities**	**Expensive,** requires **infrastructure & investment**	Adverse health effects, **highly toxic** if spilled

Future Outlook

Tailwinds

The **Inflation Reduction Act** has **$800B - $1.2T in funding,** and is directing it to Green Ammonia producers as **tax credits**

Green Ammonia **production** will be influenced by its **usage as a clean fuel**, **expanding the market** further

Over the past **10 years,** sales and **demand for Organic Foods have doubled,** thereby **increasing demand of green fertilizers**

Production of Green Ammonia Fertilizer will **reduce** farmers' **dependencies** on the **international supply chain, simplifying operations** and **maximizing use of resources**

Headwinds

Traditionally produced Ammonia is **73% cheaper** to produce

The market is **centered** around **industrial production**, but greater impact stems from **easier farmer production access**

Lack of incentives from **high costs**, creating **adoption issues** as small farm don't have the same **capital access** as large farms

Green Ammonia Fertilizer production **depends on local electricity prices** and **two New England** states rank in the top ten for the **highest energy costs**

Recent Deals

Landus Co-op & TalusAg Open Green Ammonia Facility

Event: **Landus Co-op**, an Iowan farmer-owned agricultural **cooperative**, partnered with Green Ammonia **startup TalusAg** to collectively produce **economical, yet efficient, fertilizer** for farmers

Goals: Collectively produce **economical, yet efficient, fertilizer** for farmers that is ~20%-30% less expensive than "commodity ammonia"

Influence: The U.S. **Dept. of Energy's Clean Hydrogen Production Tax Credit** (10-year cost reduction incentive) and a USDA grant worth ~$5M for construction costs (**Fertilizer Production Expansion Program**)

U.S.-Based Green Ammonia Startup Accumulates Seed Funding

Event: Ammobia, a **US-based Green Ammonia startup**, received **~$4.2M in oversubscribed seed financing** for their operations

Goals: **Advancing technology** for Green Ammonia production, applying a **1000x scale-up with the funding**, and furthering the goal of **decarbonizing agricultural fertilizer**

Influence: Their founders want to address **sustainability issues using a unique and different approach** in creating a **disruptive startup**

Healthcare: Home Healthcare

Industry Overview

- The home healthcare industry encapsulates skilled care provided in the client's home
- The market is facing an **increase in mergers and acquisitions and private equity activity**, driven primarily by consolidation efforts
- Providers are transitioning to home healthcare due to **labor shortages, tight margins, and reimbursement issues**

Course Subject Breakdown

2024 US Home Healthcare Market Outlook

7.5% CAGR → **$142.9** Billion Market Size by 2030

- The US **spends more on healthcare than any country**; affordable, high-quality care is essential

Future Outlook

Tailwinds

An aging population, with the 65+ segment to account for 20% of the US population by 2050

By 2050, the number of people 50+ with at **least one chronic disease** to rise from 71.5 million to 142.7 million, a **99.5% increase**

More **cost effective**, mean **length of stay shorter**, and **incidence of complications lower**

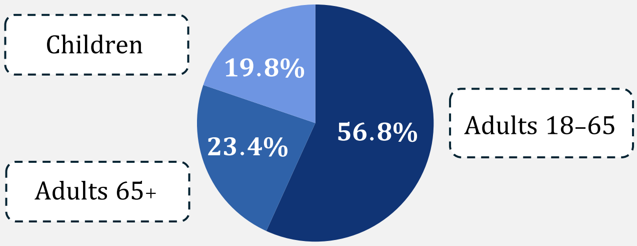
Headwinds

The home healthcare sector consists primarily of nurses, which is bracing **elevated turnover**

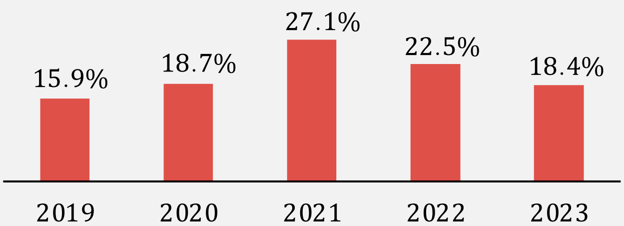
Uncertainty in **Patient-Driven Groupings Model**, **Medicare Advantage**, and **Home Health Value-Based Purchasing**

Average costs of **$4,000 – $6,000 monthly** with **substantial restrictions** on who qualifies for home healthcare under Medicare and Medicaid

Projected Age Demographics in the US, 2060



Staff Registered Nurse Turnover



Recent Deals Highlights

PathWell Home Health

Maranon Capital – $10 Million Seed Round

Trusted, Medicare-certified home healthcare for skilled nursing and therapy services

EmTech Care Labs™

Roux Institute at Northeastern University – $0.05 Million Accelerator

Care-Wallet family caregiving platform for older adults with memory disorders like dementia

Care & Help Home Care Agency

Acquired by HouseWorks – Undisclosed Amount

Leading home care provider for Philadelphia, Allentown, Harrisburg, and Pittsburgh

TMT: Disaster Relief Networks

Sector Overview

- Disaster relief networks provide victims of natural disasters with resilient means of communication and connectivity
- Advances in satellite and high-altitude platform station (HAPS) technologies will increase connectivity in remote areas
- Governments leverage disaster relief networks in short-term crisis response and long-term disaster planning & monitoring

2024 US Disaster Relief Systems Market

6.9% CAGR → $43.2 Billion Market Size by 2030

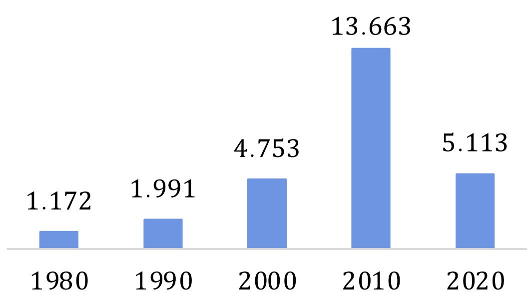
- The Northeast is experiencing the highest increase in extreme precipitation in the US, recording a **60%** surge in the frequency of heavy rainfall in 2023

Favorable Tailwinds

Climate Change and Natural Disasters

- Climate change has amplified the severity of natural disasters, displacing millions of people

North Americans Displaced by Natural Disasters per Decade, in Millions



Government Initiatives and Funding

- The Biden Administration has dedicated **$42.45 billion** to equitable broadband connectivity through the BEAD program

Urbanization and Population Density

- **80%** of the US population lives in urban areas, which are reliant on vulnerable, interconnected infrastructure

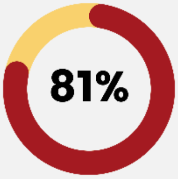
Emerging Headwinds

Weather-Induced Disruptions and Degradation

- Fog and snow impede network reliability and capacity, causing wireless signal attenuation of **over 30dB/km**

Integration and Compatibility Challenges

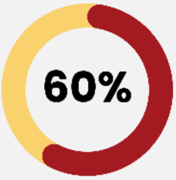
- Legacy systems and varying data needs among relief organizations create network integration challenges



81% of IT firms report data integration issues as an obstacle to digital transformation efforts

High Infrastructure and Upgrade Costs

- Reaching underserved stakeholders necessitates additional investment in infrastructure



To improve network capacity and accessibility, operating costs may increase up to **60%**

Recent Deal Highlights

COASTAL MEASURES

- Based in Durham, NH, **Coastal Measures** draws AI insights from oceanographic data to enable informed climate action
- **Coastal Measures** joined the BlueSwell Incubator program in September 2024; the firm is currently generating revenue

OWL

- Headquartered in Allentown, PA, **OWL Integrations** develops wireless communications devices for disaster relief
- **OWL Integrations** received a 100,000 EUR grant in November 2023; the firm is currently seeking venture funding

Primary Sources

Education & Finance: Online Life-Long Learning

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